

12.424

AI25BTECH11034 - Sujal Chauhan

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Question

Let $\mathbf{T} : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ be a linear map defined by

$$\mathbf{T}(x, y, z, w) = (x + z, 2x + y + 3z, 2y + 2z, w).$$

Then the rank of $\mathbf{T}$ is equal to _____.

Solution

Let $\mathbf{T}$ be the transforming matrix. Now, take any general point $\mathbf{x}$ in $\mathbb{R}^4$. We have

$$\mathbf{T} \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} x + z \\ 2x + y + 3z \\ 2y + 2z \\ w \end{pmatrix} \quad (1)$$

$$= x \begin{pmatrix} 1 \\ 2 \\ 0 \\ 0 \end{pmatrix} + y \begin{pmatrix} 0 \\ 1 \\ 2 \\ 0 \end{pmatrix} + z \begin{pmatrix} 1 \\ 3 \\ 2 \\ 0 \end{pmatrix} + w \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \quad (2)$$

Solution

Therefore, the matrix representation of $\mathbf{T}$ is

$$\mathbf{T} = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 2 & 1 & 3 & 0 \\ 0 & 2 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (3)$$

$$\boxed{\text{rank}(\mathbf{T}) = 3.} \quad (4)$$